

:- श्री शै श्री हरिदास :-

:- श्री शिवाय नमस्तुभ्यः :-

:- HTML :-

- Hyper Text markup language.

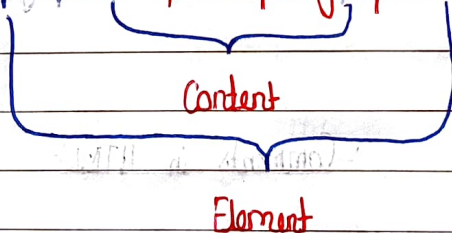
Hyper is the code that is used to structure a webpage and its content.

The component used to design the structure of websites are called HTML tags.

:- HTML Tags :-

The Container for some content or other HTML tags.

paragraph opening tag → `<p>` This is a paragraph `</p>` ← closing tag !



Ex:-

```

<!DOCTYPE html>
<html>
  <head>
    <body>
      <p> This is my book </p>
    </body>
  </html>

```

-: Basic HTML Page :-

`<!DOCTYPE html>`

tell browser you are using HTML5

`<html>`

root of an html document

`<head>`

container of metadata

`<title>My First Page</title>`

page title

`</head>`

`<body>`

contains all data rendered by the browser

`<p>Hello world</p>`

paragraph tag

`</body>`

`</html>`

-: Quick Points :-

- HTML tag is parent of head & body tag.
- Most of html elements have opening & closing tags with content in between.
- Some tags have no content in between, eg- `<br>`
- We can use inspect element / via page source to edit html.

-: Comments in HTML :-

This is part of code that should not be parsed!

`<!-- This is an HTML Command -->`

HTML is NOT case sensitive

<html> = <HTML>

<p> = <P>

<head> = <HEAD>

<body> = <BODY>

Basics HTML Tags:-

HTML Attributes:-

Attributes are used to add more information to the tag.

<html lang = "en">

↓
language

Heading Tag:-

Used to display headings in HTML.

H1 (most important)

H2

H3

H4

H5

H6 (least important)

Paragraph Tag:-

Used to add paragraphs in HTML.

<p> This is a sample paragraph </p>

Ex: <!DOCTYPE html>

<html>

<head>

<body>

<h1> my friend name is shreshth </h1>

<h2> my Instagram ID is prathamdubay2224 </h2>

<p> This friendship happend after 2 month </p>

</body>

</html>

Anchor Tag:-

Used to add links to your page

Google
↑
attribute.

 Go to Hello

Image Tag:-

Used to add images to your page.

↑
relative url.

Br Tag :-

Used to add next line (line breaks) to your page

`<br>`

Bold Italic and Underline Tags :-

Used to highlights text in your page

`<b> Bold </b>`

`<i> Italic </i>`

`<u> Underline </u>`

Big & Small Tags :-

Used to display big & small text on your page

`<big> Big </big>`

`<small> Small </small>`

Hor Tag :-

Used to display a horizontal ruler, used to separate content

`<hr>`

Subscript & Superscript Tag:-

Used to display to horizontal section used to separate content

`<sub>subscript</sub>`

H_2O

`<sup>Superscript</sup>`

$a^n + b$

Pre tag:-

Used to display text as it is (without joining spaces & next line).

`<pre>` This
is a sample
text.

`</pre>`

Page layout Techniques:-

using semantic tags for layout.
using the side tags.

`<Header>`

`<main>`

`<Footer>`

- alt tag line:-

table #alter width = 100px height = 100px { background-color: pink;

Inside Main Tag:-

- Section Tag:- For a section on your page
<section>

- Article Tag:- For an article on your page
<article>

- Aside Tag:- For content aside main content (as)
<aside>

Revisiting anchor tag:-

 Google

↑
for new tab

↑
clickable pic

Revisiting Image tag

↑
set height

↑
set width

Div Tag:-

Div is a container used for other HTML elements.

Block Elements (takes full width)

List: Block Tags:

- `<address>`
- `<code>`
- `<pre>`
- `<blockquote>`
- `<caption>`
- `<dd>`
- `<div>`
- `<dl>`
- `<dt>`
- `<fieldset>`
- `<figcaption>`
- `<figure>`
- `<form>`
- `<h1>` - `<h6>`
- `<header>`
- `<hr>`
- `<li>`
- `<main>`
- `<nav>`
- `<noscript>`
- `<ol>`
- `<p>`
- `<pre>`

Inline Element (takes width as per size)

<a> <script>

<select>

<small>

<sub>

<sup>

< + Erklärung >

<time>

 $\langle e_n \rangle$

$\langle \text{log} \rangle$

$\langle kbd \rangle$

<map>

Subject

(4)

<output>

 $\langle q_1 \rangle$

$\langle \text{amp} \rangle$

Lists in HTML :-

Lists are used to represent real life list data.

Unordered

Ordered

 Apple

 Mango

 Apple

 Mango

Tables in HTML :-

Tables are used to represent real life table data.

<tr> used to display table row

<td> used to display table data

<th> used to display table header.

<table>

<tr>

<th> Name </th>

<th> Roll no </th>

</tr>

<tr> <td> Shriji </td>

<td> 1669 </td>

</tr>

</table>

Name	Roll no
Shriji	1669

Caption in Tables:

`<caption>` Student Data `</caption>`

Student data	
EName	Roll no
Shruti	1669

thead & tbody in Tables

`<thead>` to wrap table head

`<tbody>` to wrap table body

Form in HTML:

forms are used to collect data from the users.

Eg:- sign up / login / help requests / contact me

`<form>`

form content

`</form>`

Action in Form:

Action attribute is used to define what action needs to be performed when a form is submitted.

`<form action = "/action.php">`

Form Element : Input :-

Label :-

<label for="id1">

☐

</label>

<label for="id2">

☐

</label>

Class & ID :-

<div id="id1" class="group1">

</div>

<div id="id2" class="group1">

</div>

Select :-

<select name="city" id="city">

<option value="Delhi"> Delhi </option>

<option value="Mumbai"> Delhi </option>

<option value="Bangalore"> Delhi </option>

</select>

iframe tag :-

Website inside website.

`<iframe src = "link">`

Video tag :-

`<video src = "myVid.mp4" > My Video </video>`

Attributes :-

- controls

- height

- width

- loop

- autoplay

→ CSS ←

CSS:-

Cascading style sheets

It is a language that is used to describe the style of a document.

Basic Syntax:-

```

Selector {
    color: red;
}
    
```

Annotations:
 - **Selector** points to the curly braces `{ }`.
 - **Property** points to `color`.
 - **Value** points to `red`.

Including style:-

Inline:-

`<h1 style="color:red"> Han Han Mahadev </h1>`

• Style tag:-

```

<style>
h1 {
    color:red;
}
</style>
    
```

[Refresh command
⇒ CTRL+R]

• External style sheet:-

Writing CSS in a separate document & linking it with HTML file.

Color Property:

Used to set the color of background.

color: red;

color: pink;

color: blue;

color: green;

→ text, button, link

Background Color Property

Used to set the color of background.

background-color: red;

background-color: pink;

background-color: blue;

background-color: green;

Color System:

• RGB

color: rgb(255, 0, 0);

color: rgb(0, 255, 0);

RGB = Red Green Blue

jaise koi color ko mix karna
hai jaise Ex: yellow mix color
(255, 255, 0)

RGB color set bhi karta hai
jaise Ex: red (255, 0, 0)

0 - * 255 range mai karna bhi hi chahiye hai

RGB no check in Google :- Search :- Color Picker.

- Hex (Hexadecimal)

Hex $\rightarrow$ 6 Decimal $\rightarrow$ 10.

dec means $\Rightarrow$ 10 color

color: #FF0000;

color: #00FF00;

R G B

0-255 0-255 0-255

00-ff 00-ff 00-ff

Selectors :-

- Universal Selector
* {}

[* $\Rightarrow$ asterisk]

- Element Selector
p {}

- Id Selector
#myId {}

- Class Selector
.myClass {}

Simple project

Ques 1. Create a simple div, an if id "box". Add some text content inside the div. Set its background color to blue.

$\Rightarrow$

<!DOCTYPE html>

<html lang="en">

<head>

<title> CSS Level 1 </title>

<link rel="stylesheet" href="style.css">

</head>

<body>

<div id="box">

Lorem

</div>

</body>

</html>

box {

background-color: blue;

color: white;

}

Ques 2. Create a 3 heading with h1, h2, h3.

Give them all a class "Heading" & set color "of Heading" to red.

> <!DOCTYPE html>

<html lang="en">

<head>

<title> CSS Level 1 </title>

<link rel="stylesheet" href="style.css">

</head>

<body> class="Heading">

<h1> This is my book </h1>

<h2> This is a my pen </h2>

<h3> Shri madhe Shri Haridash </h3>

</body>

</html>

Heading
~~class~~ {

color : ☐ red;

}

A₁, A₂, A₃ {

color : ☐ red;

}

Ques 3. Create a button & set its background color to:

- green using CSS stylesheet.
- blue using <style> tag.
- pink using inline style

⇒ <!DOCTYPE html>

<html lang="en">

<head>

<style>

button {

background-color : ☐ blue;

}

</style>

<link rel="stylesheet" href="style.css">

</head>

<body>

<div id="box">

Lorem

</div>

<h1 class="Heading"> Heading 1 </h1>

<h2 class = "Heading" > Heading 2 </h2>

<h3 class = "Heading" > Heading 3 </h3>

<button style = "background-color: # ;" > Click me ! </button>

</body>

</html>

button {

background-color: # green ;

}

CSS

-: Text Properties :-

text-align

support CSS3

text-align : left / right / center / start / end

English	left = start
Hindi	left = start
Urdu	left = end, right = start

-: Example :-

<DOCTYPE html>

<html lang = "en">

<head>

<title> CSS level 1 </title>

<link rel = "stylesheet" href = "style.css">

</head>

<body>

<h1> Heading </h1>

<p> Lorem * 10 </p>

</body>

</html>

HTML

CSS

p {

text-align : left;

}

p {

text-align : right;

}

p {

text-align : center;

}

h1 {

text-align : right;

}

Text Properties :-

text-decoration

text-decoration : underline / overline / line-through

CSS

h1 {

text-align : right;

text-decoration : underline;

}

h1 {

text-align : right;

text-decoration : overline;

}

h1 {

text-align : right;

text-decoration : line-through;

}

h1 {

text-align : right;

text-decoration : none;

}

h1 {

text-align : right;

text-decoration : blue wavy underline;

}

Text decoration style:-

text-decoration-style: solid;
text-decoration-style: double;
text-decoration-style: dotted;
text-decoration-style: dashed;
text-decoration-style: wavy;

Text properties:-

font-weight

font-weight: normal / bold / bolder / lighter;
font-weight: 100 - 900

lightest shade - 100
darkest shade - 900

`<h1 id="h1">heading 1 </h1>`
`<h1 id="h2">heading 2 </h2>`
`<h1 id="h3">heading 3 </h1>`
`<h1 id="h4">heading 4 </h1>`

HTML

CSS

#h1 {	#h2 {	#h3 {
font-weight: normal;	font-weight: bold;	font-weight: bolder;
100	700	800

#h4 {
font-weight: lighter;
900

Font Properties :-

font-family / style

fall back mechanisms

font-family: serif

font-family: serif, sans-serif

CS3

ps

font-family: serif;

↓

Two types of font families :-

- Generic font families
- Specific font families

Units in CSS

Absolute

↑ { relation of size }

pixels (px)

96 px = 1 inch = 2.54 cm

font-size: 2px;

one {

font-size: 25px;

}

CSS

```
<p id="one">para1</p>
<p id="two">para2</p>
<p id="three">para3</p>
<p id="four">para4</p>
```

HTML

three {

font-size: 10px;

}

-: Text Properties :-

Line-Height

line-height : 2px

line-height : normal

-: CSS :-

#one {

font-size : 25px;

line-height : 50px;

-: Text Properties :-

Text-transform

text-transform : uppercase / lowercase / capitalize / none

-: CSS :-

#one {

font-size : 25px;

line-height : normal;

text-transform : uppercase;

-: Practice :-

Q1. Create a heading centered on the page with all of its text capitalize by default

⇒ <DOCTYPE html>

<html lang="en">

<head>

<title> CSS level 1 </title>

<link rel="stylesheet" href="style.css">

</head>

<body>

<h1> Radhe </h1>

</body>

</html>

-: CSS :-

h1 {

text-align : center;

text-transform : capitalize;

}

Q2. Set the font family of all the content in the document to "Times New Roman".

⇒ HTML CODE SAME Q-1

∴ CSS: Universal Selector

* {

font-family: "Times New Roman";

}

Minute change

Q3. Create one div inside another div.

Set id & text "outer" for the first one & "inner" for the second one. Set the outer div text size to 25px & inner div text size to 10px.

<h1> श्री राधे </h1>

<div id="outer">

<div id="inner">

∴ CSS:

#outer {

font-size: 25px;

}

</div>

</body>

</html>

#inner {

font-size: 10px;

}

∴ BOX MODEL IN CSS:

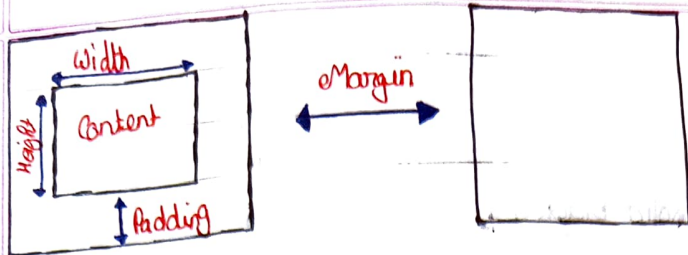
• Height

• Padding

• Width

• Margins

• Border



Border.

Height:-

By default, it sets the content area height of the element.

div {

height: 50px;

}

div {

height: 200px;

background-color: pink;

}

Width:-

By default, it sets the content area width of the element.

div {

width: 50px;

}

div {

height: 100px;

width: 100px;

background-color: pink;

}

Border:-

Used to set an element's border

border-width: 2px;

border-style: solid / dotted / dashed

border-color: black;

div {

height: 100px;

width: 100px;

background-color: pink;

border-width: 2px;

border-style: solid;

border-color: brown;

}

Border :-

Border

border: 2px solid black ;
 ↑ ↑ ↑
 width style color

border-width: 2px ;

border-style: solid ;

border-color: brown ;

⇒

border: 2px solid brown ;

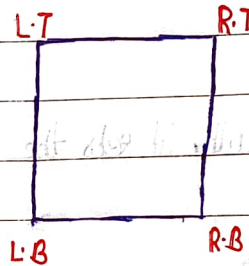
text-align: right ;

Border :-

Used to round the corners of an element's outer border edge

border-radius: 10px ;

border-radius: 50% ;



:- CODE :-

<DOCTYPE html>

HTML

<html lang="en">

<head>

<title> ~~document~~ CSS level 2 </title>

<link rel="stylesheet" href="style.css">

</head>

<body>

<div> div1 </div>

<div> div2 </div>

<div> div3 </div>

</body>

</html>

div {

height: 100px;

width: 100px;

background-color: pink;

border: 2px solid brown;

text-align: right;

border-radius: 25px;

}



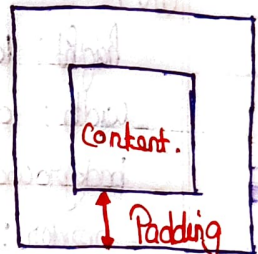
• Padding:-

• Padding-left

• Padding-right

• Padding-top

• Padding-bottom



CSS

div {

height: 100px;

width: 100px;

background-color: pink;

border: 2px solid brown;

padding-left: 25px;

padding-right: 25px;

}

Padding shorthand:-

Shorthand

padding: 5px;

padding: 1px 2px 3px 4px;

top/right/bottom/left → clockwise

CSS

div {

height: 100px;

width: 100px;

background-color: #pink;

border: 2px solid #brown;

padding: 1px 2px 3px 4px;

}

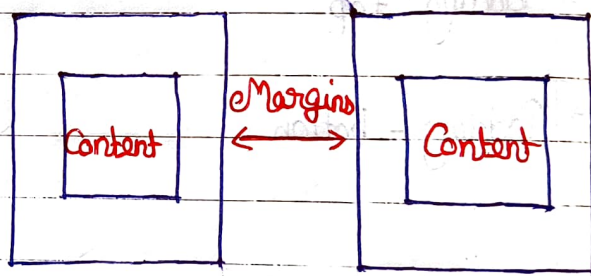
• Margins:-

→ margins - right

→ margins - left

→ margins - top

→ margins - bottom



CSS

div {

height: 100px;

width: 100px;

background-color: #pink;

border: 2px solid #brown;

padding: 1px 2px 3px 4px;

margins-top: 50px;

}

Margins Shorthand:

Shorthand:

margins: 50px;

margins: 1px 2px 3px 4px;

top / right / bottom / left → clockwise

div {

height: 100px;

width: 100px;

background-color: pink;

border: 2px solid #brown;

padding: 25px;

margins: 25px;

Practice:-

- Q1. Create a div with height & width of 100px.
Set its background color to green & the border radius to 50px.

<DOCTYPE html>

<html lang="en">

<head>

<title> CSS Label 2 </title>

<link rel="stylesheet" href="stylesheet.css">

</head>

<body>

<div> </div>

</body>

</html>

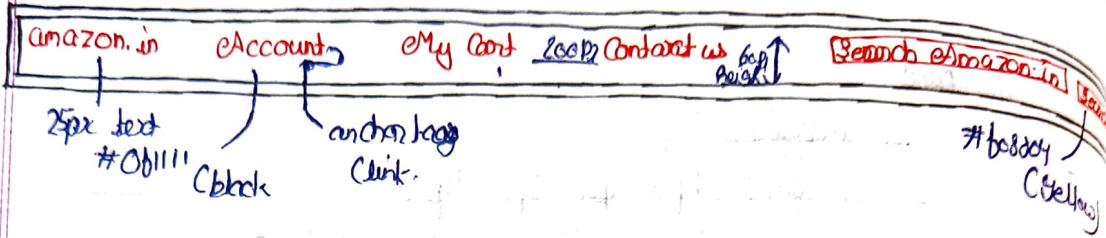
height: 100px;

width: 100px;

background-color: green;

border-radius: 50px;

Ques 2: Create a navbar.



HTML

```
<!DOCTYPE HTML>
<HTML lang = en>
<Head>
  <title> CSS level 2 X /title>
  <link rel = "stylesheet" href = "Style.css">
</Head>
<body>
  <div id = "navbar">
    <a id = "logo"> amazon.in </a>
    <a> Account </a>
    <a> My Cart </a>
    <a> Contact Us </a>
    <input placeholder = "Search Amazon.in">
    <button> Search </button>
  </div>
</body>
</html>
```

CSS

```
* {
  padding : 0;
  margin : 0;
  color : white;
}
```

navbar {

height: 60px;

background-color: #0f1111;

}

button {

background-color: #f08804;

}

logo {

color: #f08804;

font-size: 25px;

}

text-decoration: none;

Display Property:

display: inline / block / inline-block / none.

- inline: Takes only the space required by the elements. (no margins/padding)
- block: Takes full space available in width.
- inline-block: Similarly to inline but we can set margins & padding.
- none: To remove element from document flow.

Visibility:

visibility: hidden;

Altair When visibility is set to be none, space for the element is reserved. But for display set to none, no space is reserved or blocked for the element.

Alpha Channel:

opacity (0 to 1).

• RGBA:

color: rgba(255, 0, 0, 0.5);

color: rgba(255, 0, 0, 1);

Practice:

Ques 1:- Create a webpage layout with a header, a footer & a content area containing 3 divs.

Set the height & width of divs to 100px.

(Add the navbar in the header).

⇒ <!DOCTYPE html>

<html lang="en">

<head>

<title> CSS Level 2 </title>

<link rel="stylesheet" href="style.css">

</head>

<body>

<header>

<div id="navbar">

" " "

" " "

</div>

</header>

<div id="content">

<div class="contentBox"></div>

<div class="contentBar"></div>

Page No. _____
Date ____/____/____

```
<div class="contentBox"></div>  
</div>  
<footer> footer </footer>
```

```
</body>  
</html>
```

Copy Old navbar :-

:- CSS :-

```
.contentBox {  
    height: 100px;  
    width: 100px;  
}
```

Ques 2: Add border to all the divs :-

⇒

```
.contentBox {  
    height: 100px;  
    width: 100px;  
    border: 2px solid black;  
}
```

:- CSS :-

Ques 3: Add a different background color to each div with an opacity of 0.5.

⇒

:- HTML :-

```
<!DOCTYPE>  
<html lang="en">  
<head>
```

```
    <title CSS Revol 2 </title>  
    <link rel="stylesheet" href="style.css">  
</head>
```

<body>

<Header>

<div id="navbar">

" "

" "

" "

</div>

</Header>

<div id="content">

<div class="contentBox" style="background-color: rgba(255, 0, 0, 0.5);">

</div>

<div class="contentBox" style="background-color: rgba(0, 255, 0, 0.5);">

</div>

<div class="contentBox" style="background-color: rgba(255, 0, 0, 0.5);">

</div>

<Footer>

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</Footer>

</body>

</html>

Copy of navbar + CSS :-

.contentBox {

height: 100px

width: 100px

border: 2px solid black;

display: inline-block;

}

```

body {
background-color: #0b1111;
color: white;
text-align: center;
}

```

Ques 4:- Give the content area an appropriate height.

```

⇒ .content Box {
height: 100px;
width: 100px;
border: 2px solid black;
display: inline-block;
}

```

= CSS :-

```

# Content {
height: 500px;
}

```

```

body {
background-color: #0b1111;
color: white;
text-align: center;
}

```

Units of CSS :-

Relative

%

em

rem

Percentage (%) :-

It is often used to define a size as relative to an element's parent object.

width : 33% ;

margin-left : 50% ;

∴ Example :

HTML

<!DOCTYPE html>

<html lang="en">

<head>

<title> CSS Level 3 </title>

<link rel="stylesheet" href="style.css">

</head>

<body>

<div id="box1">

<div id="box2"> box 2 </div>

</div>

</body>

</html>

box1 {

height : 100 px ;

width : 200 px ;

background-color : # green ;

}

:- CSS :-

box2 {

height : 50 px ;

width : 30 % ;

background-color : # Yellow ;

margin-left : 10 % ;

}

\$:- (Quantifier) :- The n\$ quantifier matches any string with n at the end of it.